Daric Security MCU

CR7SE01

ARM M7 MCU and integrated Secure Element
TSMC 22nm ULL Process Node

Data Sheet – Advanced Information

Features

- **Powerful ARM M7 CPU**
  - 800MHz CPU/400MHz AXI
  - FPU, double precision
  - MPU, 16 entries
  - Icache 16kB, Dcache 16kB
  - iTCM 256kB, dTCM 64kB
- RISC-V CPU
  - 400MHz CPU
  - MMU
  - Icache 16kB, Dcache 16kB
- **Large high-speed memory system**
  - 4MByte embedded NVM
  - RRAM robust against physical attack, aging, & environment
  - 2MByte main SRAM
  - Separate buffer RAM for I/O
  - 2x QSPI for off-chip memory
- **Flexible I/O set**
  - I2S, I2C
  - Q/SPIM, SPIS
  - Camera IF
  - SD Host
  - UART
  - USB2.0
  - QSPI Flash controller
  - Aux ADC/PWM DAC
  - GPIO/GPINT
- **Embedded Cryptographic Engine**
  - Isolated from CPU bus
  - Asymmetric cryptography
    - ECDSA, Schnorr, EdDSA and ECDH (secp256k1, P-256/384, ed25519, Curve25519)
    - RSA, up to 8192 bit
  - Symmetric cryptography
    - AES
  - Homomorphic Encryption
    - Paillier cryptosystem
  - Hashes
    - SHA2, SHA3/Keccak, RIPEMD160, BLAKE2/3
  - Authentication codes
    - HMAC 256/512
- High Speed **TRNG**
- **Monotonic counters**
- **Logical security**
  - Boot / lifecycle protection
  - Test mode sealing / debug mode sealing
  - Crypto engine isolated from CPU
  - Detailed memory protection/privilege
- **Physical Countermeasures**
  - Active mesh
  - Glue cell network
  - Voltage glitch detection
  - Chip decap sensing
  - Thermal attack
  - Memory fault detection
  - False injection detection
  - Side channel protection, clock jitter, power balancing
  - Redundant key registers
  - Puzzle routing
  - Secret zeroization
- **Integrated PMU**
  - Operation from single supply
- **Software Development Kit**
- **Optional packages**
  - 188BGA 8mm X 8mm for MCU+SE function
  - 71 Ball WLCSP for limited function
  - KGD
- **Operational temperature**
  - -20C to 85C ambient

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1 Description

Daric is a novel integrated general purpose Microcontroller Unit (MCU) and Secure Element (SE), all under a common umbrella of logical countermeasures (LCMs) and physical countermeasures (PCMs). The entire chip has the physical security of a typical SE, but unlike standalone SE chips, also has the computing power of a powerful general purpose MCU. By using novel RRAM and 22nm ULL technology, speed, power, memory size, and security are all markedly enhanced. Key Features include:

Integrated MCU and SE

- Integrating both functions on a single die reduces cost, power, and area, and enhances security.
- Shielded MCU can secure important functions outside of SE, such as user display and input, which otherwise are typically executed in a general purpose MCU outside of the shield of PCMs.
- Most advanced MCU in M-series (M7) with advanced features such as cache, TCM, floating point unit, AXI bus fabric.
- Alternate boot mode into RiscV system with MMU (separate SKU option).

Code and NV data are stored in Resistive RAM, with many advantages over legacy floating gate flash.

- Not readable by simple physical attack such as passive voltage contrast or electron microscopy.
- Long retention (does not leak charge continuously).
- Robust against magnetic fields and high energy particles.
- High memory density enabling larger code/data space than comparable MCUs.
- Integrated with small 22nm ULL process (which floating gate cannot be).

The advanced 22nm ULL process has many advantages in the context of secure MCU:

- Low cost, power, size
- High speed operation (800MHz)
- Higher memory sizes due to high NVM and SRAM density
- Small feature size is naturally more resistant to physical attack, FIB, probe, and inspection.
- Countermeasures are also more effective due to high feature density and lower emissions.

Advanced Cryptographic Architecture

- Support for modern curves and functions with known performance and security advantages
- High throughput and low latency
- Atomic functions may be chained together directly in SE
    - Has the system isolation and simplicity of a direct hardware SE
- Alternately, the MCU may access atomic functions
    - Thus entire chip, including MCU, may be used as a flexible, physically shielded, “super SE” for cryptographic innovation

Open Development

- Supported with devkit and open development tools
- Open source strategy unlike typical banking/payment SE’s

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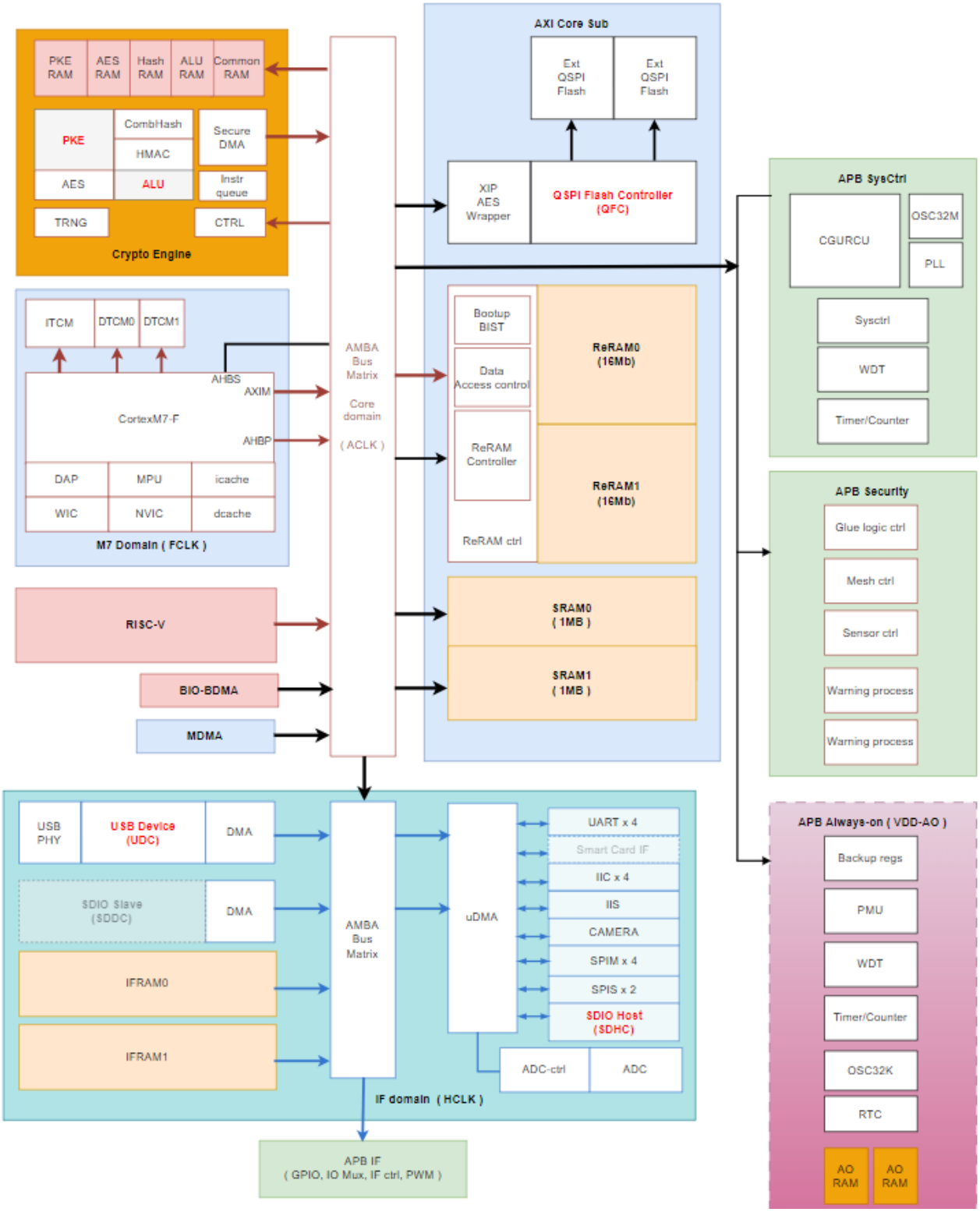
2 Block Diagram



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3 Revision History

Rev. #	Date	Changes
.1	June 14, 2022	Initial draft
.2	Feb 11, 2023	Updated Crypto Features, RiscV Option
3	March 11, 2025	Overhauled to reflect latest architecture, features and performance

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